

AI KOL — Weekly Report

Sofia · week ending 20 August 2026 · 31 commits

1. WHAT I DID THIS WEEK

- **Voice quality.** Her voice was a zero-shot clone from 6.5 s of audio. Added the trained RVC model as a timbre pass, served resident so it costs ~1 s instead of 6.4 s.
- **Response latency and continuity.** Answers publish as soon as the first audio exists, and short answers render as one continuous clip.
- **Humour training.** Took two labelled HuggingFace datasets (SMILE-Next, conversations_humor), rewrote 283 Q&A pairs into her voice, retrained the model.
- **Voice options.** Added a voice picker to the demo with 3 licensed candidate timbres, for a possible rebrand. Declined to clone the named idol voices — real people, no consent, and against the project's own rights policy.
- **Test tooling.** A 12-question probe aimed at known failure modes, so changes are checked the same way each time.

2. IMPROVED ON LAST WEEK

Measure	Before	Now	Measure	Before	Now
Voice identity vs reference	0.894	0.929	Probe failures (of 12)	4–6	2
Pauses inside one answer	1–6	0	Ends on a question back	2 / 12	0 / 12
Audio clips per answer	7	1	Invents a fake bereavement	sometimes	not seen
Render vs speech (RTF)	1.23	0.83	Generic comfort replies	present	gone

Correction worth recording: the metric used all week to measure humour was invalid. Checked against two labelled datasets, it fires on 0.3% of lines that actually made a person laugh, and slightly more often on deliberately flat answers than funny ones. Whether she is funnier is therefore a human judgement, not a measured one. The gains above use other, validated measures.

3. CURRENT PROBLEM — MAKING HER ANSWER MORE INTELLIGENTLY

- **She invents specifics.** Asked for her usual café she names "Café Buena Vista in Little Havana, owner Maria." It does not exist. Humour training worsened this: the confident exact detail that lands a joke also makes a fake business sound real.
- **Biography drifts.** After training, 1 answer in 6 about where she lives is wrong (she claims to have lived in Taiwan; she lives in Miami).
- **Root cause:** retrieval supplies her real life to the model, but *nothing checks the answer against it*. There is no step that asks "is this fact in her file?"
- **Proposed fix:** verify named entities in each reply against `life.json`, and refuse or regenerate when a place, person or business appears that is not hers. This is code, not prompting — every prompt-based fix attempted this week failed to move behaviour.

4. HOW RAG WORKS HERE

- **Store:** `life.json`, ~130 hand-written facts in typed sections — stories, biography, opinions, people, ongoing threads, knowledge.
- **Retrieval:** keyword overlap, not embeddings. The choice is deliberate: ranking ~50 short strings does not justify an embedding model inside the response budget.
- **Anchors bypass retrieval.** Five core facts are injected every turn. Sampling a fact means that on turns it is not drawn she has no evidence for her own biography — which is how she once denied a trip written in her own file.
- **Per-character alias map.** "Taiwan" matches nothing, because every story names Taipei, Tainan or Jiufen. The alias map expands the query so a place is found under all its names.
- **Conditional sections.** Biography and factual knowledge load only when the question reaches for them; unused reference costs tokens in and seconds of speech out.
- **Also used in training.** Retrieval runs during dataset generation too, so the model is fitted on answers already grounded in her real life.

All figures reproducible via `tools/llm_train/probe_answers.py` and the voice evaluation scripts.